macro

Veda A-106

Security Audit

October 1st, 2026
Version 1.0.0

Presented by 0xMacro

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Introduction

This document includes the results of the security audit for Veda Labs's smart contract code as found in the section titled ‘Source Code’. The audit was performed by the Macro security team on October 1st, 2026.

The purpose of this audit is to review the source code of certain Veda Labs Solidity contracts, and provide feedback on the design, architecture, and quality of the source code with an emphasis on validating the correctness and security of the software in its entirety.

Disclaimer: While Macro’s review is comprehensive and has surfaced some changes that should be made to the source code, this audit should not solely be relied upon for security, as no single audit is guaranteed to catch all possible bugs.

Overall Assessment

No issues or discrepancies were identified within the defined scope of this audit.

Specification

Our understanding of the specification was based on the following sources:

- Discussions with the Veda Labs team.
- Available documentation in the repository, including the description of each described Pull Request.

Trust Model, Assumptions, and Accepted Risks (TMAAR)

The changes in scope add two token issuers as new trusted parties for the vaults that use them.

Ondo (USDY and rUSDY issuer). Vaults using the Ondo leaves rely on Ondo for:

- keeping the vault registered in `OndoIDRegistry` and off the USDY blocklist and sanctions list
- keeping redemptions unpaused and USDC an accepted token
- its oracle, fees, price floor and rate limits (10 million USD of redemptions per 24 hours per user ID, shared by every address under that ID, and 15 million USD across all Ondo users), all changeable at any time
- the usual issuer powers over the tokens: both are upgradeable and rUSDY exposes an admin `burnShares()`

Any of the first three can delay or block the exit from USDY and rUSDY, but cannot redirect a payout, which always goes to the calling vault.

Hastra (wYLDS and PRIME issuer). Vaults using the Hastra leaves or `HastraPrimeBufferHelper` rely on Hastra for:

- keeping wYLDS and PRIME unpaused and the vault unfrozen; both powers are held by EOAs and take effect immediately
- keeping the PRIME NAV fresh: a report older than 3600 seconds reverts every PRIME conversion and, with the buffer helper set, every USDC deposit through the teller until the helper is unset
- settling exits from wYLDS, which are asynchronous (one pending redemption per address) and settled all or nothing by Hastra's operator from its redeemVault (about 98 thousand USDC at review time), shared by every Veda vault entering PRIME this way
- upgrades and redeemVault changes, which go through a 1 day timelock

Source Code

The following source code was reviewed during the audit:

branch and commit.

Repository: boring-vault-private

PR 153, Ondo USDY decoder

Branch `willtobin/scfeat-43-ondo-usdy-instant-mintredeem-decoder` at commit `38c2c22a9176d109be2a32efdfb5b2952ab3a9cc`

PR 123, Hastra PRIME decoder and buffer helper

Branch `will/scfeat-39-hastra-prime-decoder` at commit `40627627d0e68e8a5361f82e6fddaecedd357876`

PR 121, CCIP decoder V0.2 (GenericExtraArgsV2)

Branch `feat/ccip-decoder-extra-args-v2` at commit `5c103db17ed0cd8820e3894b69609041aef8dcff`

PR 156, Uniswap V3 SwapRouter02 adapter

Branch `feat/add-swaprouter02-adapter` at commit `fb393d10bf32416c4d0b33a7155050f7a358cff1`

Specifically, we audited the changes each pull request makes to the following files, at the commit of that pull request. Files added by a pull request were reviewed in full.

PR 153 (commit `38c2c22a9176d109be2a32efdfb5b2952ab3a9cc`):

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/OndoDecoderAndSanitizer.sol	`c99279bbd69319af980718a7a3c898c41c1fcde09ddb55705f6c04908ba6183f`

PR 123 (commit `40627627d0e68e8a5361f82e6fddaecedd357876`):

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/HastraPrimeDecoderAndSanitizer.sol	`eb665927a9b4510e7724698bb2e713ae1c0ad8897f697f3aed7ea5794bb8cdd7`
src/base/DecodersAndSanitizers/HastraPrimeFullDecoderAndSanitizer.sol	`8bbdc3b9f846d07eb57913129f6b6ec820b43cae8f34ff3de8c3423bf83712c9`
src/base/Roles/HastraPrimeBufferHelper.sol	`83a606e2d1d8312f89d79e7c4c779c44916c4c80e5c0b0ae1b28d49545a8b0bd`

PR 121 (commit `5c103db17ed0cd8820e3894b69609041aef8dcff`):

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/CCIPDecoderAndSanitizer.sol	`b1a08a009752ba92aa86c513e8fd06cb7845354b5a993549b983d1d14f7e620a`
src/interfaces/DecoderCustomTypes.sol	`0c946217bdea371b013aa30276f1980fe5d3dfccd4ab75013858414dd583c4a4`
script/GigaDeployDecoderAndSanitizer.s.sol	`8c9458ca0eaa2f65cbd53c2a69f48795d48081b5bbc53249f8c428cc160dfa17`
script/DeployDecodersAndSanitizersWithNoConstructorArgs.s.sol	`0a050ee79fead9a7f45dfbf997bbbfbd1e390e04cd835dfd4259565d01bcac31`

PR 156 (commit `fb393d10bf32416c4d0b33a7155050f7a358cff1`):

Source Code	SHA256
src/base/Periphery/adapters/UniswapV3SwapRouter02Adapter.sol	`4ea5ebcb2bf65a91ab840936c7f0773e9be8d4fca405de3e771418abf45decd0`

Security Level Reference

We quantify issues in three parts:

1. The high/medium/low/spec-breaking **impact** of the issue:
    - How bad things can get (for a vulnerability)
    - The significance of an improvement (for a code quality issue)
    - The amount of gas saved (for a gas optimization)
2. The high/medium/low **likelihood** of the issue:
    - How likely is the issue to occur (for a vulnerability)
3. The overall critical/high/medium/low **severity** of the issue.

This third part – the severity level – is a summary of how much consideration the client should give to fixing the issue. We assign severity according to the table of guidelines below:

Severity	Description
(C-x) Critical	We recommend the client **must** fix the issue, no matter what, because not fixing would mean **significant funds/assets WILL be lost.**
(H-x) High	We recommend the client **must** address the issue, no matter what, because not fixing would be very bad, *or* some funds/assets will be lost, *or* the code's behavior is against the provided spec.
(M-x) Medium	We recommend the client to **seriously consider** fixing the issue, as the implications of not fixing the issue are severe enough to impact the project significantly, albiet not in an existential manner.
(L-x) Low	The risk is small, unlikely, or may not relevant to the project in a meaningful way. Whether or not the project wants to develop a fix is up to the goals and needs of the project.
(Q-x) Code Quality	The issue identified does not pose any obvious risk, but fixing could improve overall code quality, on-chain composability, developer ergonomics, or even certain aspects of protocol design.
(I-x) Informational	Warnings and things to keep in mind when operating the protocol. No immediate action required.
(G-x) Gas Optimizations	The presented optimization suggestion would save an amount of gas significant enough, in our opinion, to be worth the development cost of implementing it.

Issue Descriptions and Recommendations

There were no significant issues found in the code reviewed.

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The scope of this report and review is limited to a review of only the code presented by the Veda Labs team and only the source code Macro notes as being within the scope of Macro's review within this report. This report does not include an audit of the deployment scripts used to deploy the Solidity contracts in the repository corresponding to this audit. Specifically, for the avoidance of doubt, this report does not constitute investment advice, is not intended to be relied upon as investment advice, is not an endorsement of this project or team, and it is not a guarantee as to the absolute security of the project. In this report you may through hypertext or other computer links, gain access to websites operated by persons other than Macro. Such hyperlinks are provided for your reference and convenience only, and are the exclusive responsibility of such websites' owners. You agree that Macro is not responsible for the content or operation of such websites, and that Macro shall have no liability to your or any other person or entity for the use of third party websites. Macro assumes no responsibility for the use of third party software and shall have no liability whatsoever to any person or entity for the accuracy or completeness of any outcome generated by such software.